

Lecture 3

Convolutional Networks and Transfer Learning

Advanced Topics in Artificial Intelligence · UFABC

Part 1 — Computer Vision

How to represent images and what tasks to solve with DL.

Tensors and images

Tensor = N-dimensional array of numbers:

rank 0	rank 1	rank 2	rank 3	rank 4
escalar	vetor	matriz	cubo	rank-4
				
42 um número	[a, b, c, d] lista de números	imagem cinza $H \times W$	imagem RGB $H \times W \times 3$	batch RGB $N \times H \times W \times 3$

Grayscale image

Matrix $H \times W$ — each pixel: 0–255

Color image (RGB)

Tensor $H \times W \times 3$ — 3 channels (R, G, B)

Batch of images

Tensor $N \times H \times W \times 3$

Computer vision tasks

Classificação



qual é a classe?

Classification

One class per image

Localização



classe + caixa

Detection

Multiple bounding boxes + classes

Detecção de objetos



várias classes/caixas

Segmentation

Class per pixel

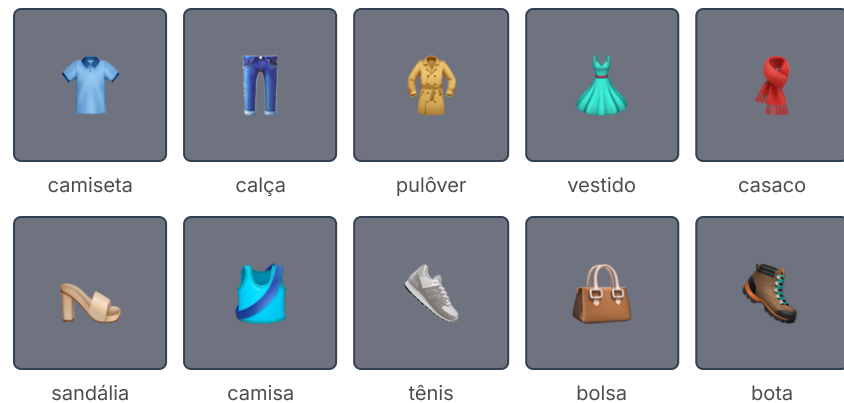
Segmentação semântica



classe por pixel

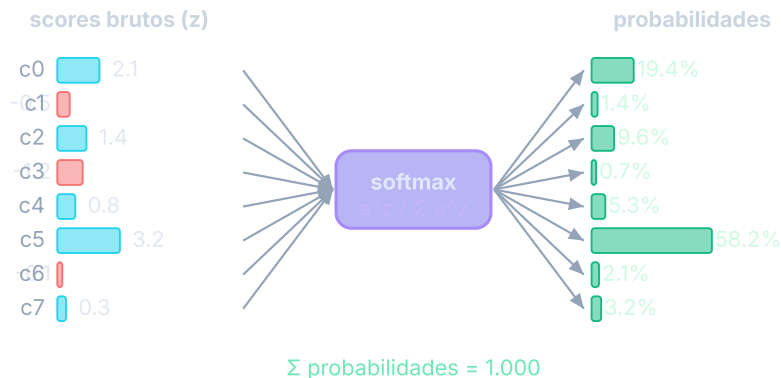
Multiclass classification — softmax

Fashion-MNIST: 70k images 28×28, 10 categories.



For N classes, the output layer uses **softmax**:

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}}$$



Outputs $\in (0, 1)$ summing to exactly 1.

Matching output to loss

Output variable	Output layer	Loss (Keras)
Real number	linear (1 neuron)	<code>mse</code>
Binary probability	sigmoid (1 neuron)	<code>binary_crossentropy</code>
Probability vector	softmax (N neurons)	<code>categorical_crossentropy</code>
Same, labels as integers	softmax (N neurons)	<code>sparse_categorical_crossentropy</code>

⚠ Using the wrong loss for the label type is one of the most common sources of bugs.

Baseline: dense network on Fashion-MNIST

Keras

```
inp = keras.Input(shape=(28, 28))
x = keras.layers.Flatten()(inp)
x = keras.layers.Dense(128, 'relu')(x)
x = keras.layers.Dropout(0.3)(x)
x = keras.layers.Dense(64, 'relu')(x)
out = keras.layers.Dense(10, 'softmax')(x)
model = keras.Model(inp, out)
model.compile(
    loss='sparse_categorical_crossentropy',
    optimizer='adam', metrics=['accuracy'])
```

PyTorch

```
model = nn.Sequential(
    nn.Flatten(),
    nn.Linear(784, 128), nn.ReLU(),
    nn.Dropout(0.3),
    nn.Linear(128, 64), nn.ReLU(),
    nn.Linear(64, 10),
)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters())
# CrossEntropyLoss already includes softmax!
```

Baseline of ~88% with dense layers. Can we do better?

Part 2 — Why dense layers are not enough

The problem of flattening images.

The problem with Flatten

- Color image from a smartphone: **$3024 \times 3024 \times 3$ pixels**
- Flatten and connect to a layer of 100 neurons $\rightarrow \approx$ **2.7 billion parameters**
- Computationally infeasible, requires enormous amounts of data, prone to overfitting

By flattening, we lose the **spatial structure** — neighboring pixels carry local information that the dense layer ignores.

If a pattern appears at different positions in the image, the dense network needs to **re-learn it** for each position. Convolutional filters **learn once and reuse** at any position.

Flatten: what is lost

4×4 image (original)

A	B	.	.
C	.	.	.
.	.	.	.
.	.	.	.

A-B: neighbors → ✓

A-C: neighbors ↓

→
Flatten

1×16 vector (after Flatten)

A	B	.	.	C	.	.	.	.	.	.	.	.	.	.	.
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

A-B: still neighbors ✓

A-C: now 4 positions apart ✗

Parameter explosion

$224 \times 224 \times 3 \rightarrow 128$ neurons = **19M params** in that layer alone

Loses invariance

Cat in the corner $\neq$ cat in the center — the network treats them as distinct inputs

Loses 2D adjacency

Vertical neighbors end up *width* positions apart in the vector

Part 3 — Convolutional Filters

How to detect visual patterns efficiently.

The convolutional filter

- A **filter** is a small matrix of numbers (e.g.: 3×3)
- Sliding over the image, it detects a type of visual pattern
- A **convolutional layer** has multiple filters; each one learns a different pattern

Vertical edge:	Horizontal edge:
1 0 -1	1 1 1
1 0 -1	0 0 0
1 0 -1	-1 -1 -1

Analogy with dense neuron:

- Dense neuron → connected to **all** pixels
- Convolutional filter → connected to a **local window** and slides with the **same weights**

Benefits:

- **Far fewer parameters**
- Preserves **spatial adjacency**
- **Translation invariance** — detects the same pattern at any position

The convolution operation

Steps:

1. Position the filter over a window of the image
2. Multiply element-wise and sum $\rightarrow$ a scalar
3. Slide (stride) and repeat
4. Result: a **feature map**
5. Apply ReLU for non-linearity

```
3x3 window:  Filter (vertical edge):  
1  2  3      1  0 -1  
4  5  6  ×   1  0 -1  = (1+4+7)-(3+6+9) = -6  
7  8  9      1  0 -1
```

High value in the feature map = filter **detected** the pattern in that region.

By stacking layers:

- Layer 1 $\rightarrow$ edges and simple textures
- Layer 2 $\rightarrow$ corners, curves
- Layer 3+ $\rightarrow$ object parts, complete objects

Convolution step by step

6×6 input (excerpt) — 3×3 filter slides with stride 1:

```

┌───┬───┬───┬───┐
│ 1 │ 2 │ 3 │ 0 1 │
├───┼───┼───┼───┤
│ 4 │ 5 │ 6 │ 1 0 │ → position (0,0) → -6
├───┼───┼───┼───┤
│ 7 │ 8 │ 9 │ 0 1 │
└───┴───┴───┴───┘

```

```

. ┌───┬───┬───┬───┐ .
. │ 2 │ 3 │ 0 │ 1 │ → position (0,1) → 3
. │ 5 │ 6 │ 1 │ 0 │
. │ 8 │ 9 │ 0 │ 1 │
. └───┴───┴───┴───┘ .

```

Filter (vertical edge):

```

1  0 -1
1  0 -1
1  0 -1

```

Resulting feature map (4×4):

```

-6  3  ...
...  ...

```

Output size: $(H - F + 1) \times (W - F + 1)$

6×6 input, 3×3 filter → **4×4** output

With **padding=1** → **6×6** output (same resolution)

What filters detect

Vertical edge

1	0	-1
1	0	-1
1	0	-1

High response where intensity changes from left to right

Horizontal edge

1	1	1
0	0	0
-1	-1	-1

High response where intensity changes from top to bottom

Blur (smoothing)

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Average of neighbors → smooths noise and details

In CNNs, **filter values are not hand-crafted** — they are **learned by backprop**. The network discovers on its own which patterns are useful for the task. Deep layers combine edges → corners → parts → objects.

Parameters of a convolutional layer

- **Filter size:** 3×3 or 5×5 (3×3 preferred)
- **Number of filters:** 32, 64, 128... per layer
- **Stride:** sliding step (1 or 2)
- **Padding:** add zeros at the borders to preserve resolution

Parameters of 32 filters 3×3 with 3 channels:

$$32 \times (3 \times 3 \times 3 + 1) = 896$$

Equivalent dense layer ($224 \times 224 \times 3 \rightarrow 32$):

≈ 4.8 million parameters

Part 4 — Pooling

Reducing dimensions without losing the essentials.

Max Pooling

- Divides the feature map into windows (e.g.: 2×2)
- Retains only the **maximum value** of each window
- Reduces spatial resolution by half
- Fewer parameters in subsequent layers

Intuition: if a feature exists **anywhere** in the window, max pooling preserves it. Provides robustness to small shifts.

Feature map (4×4): MaxPool 2×2 :

1	3		2	4		3	4
5	6		1	2	→	6	5
-----+-----							
7	2		3	1		7	5
4	1		5	2			

Max Pooling vs Average Pooling

Max Pooling — preserves the *presence* of the feature

Feature map (4×4):

1	3	2	4
5	6	1	2
7	2	3	1
4	1	5	2

→ MaxPool 2×2:

6	4
7	5

Maximum of each window: "does this pattern **exist** here?"

MaxPool — standard in vision CNNs; detects feature presence

Average Pooling — preserves the *average intensity*

Feature map (4×4):

1	3	2	4
5	6	1	2
7	2	3	1
4	1	5	2

→ AvgPool 2×2:

3.75	2.25
3.5	2.75

Average of each window: "what is the **typical** intensity of this region?"

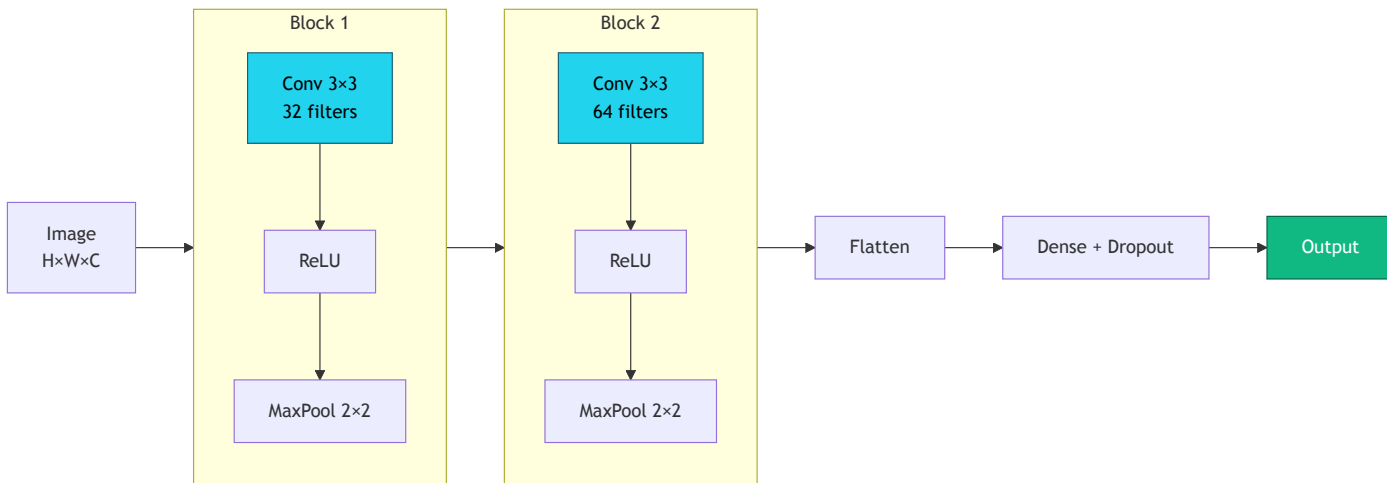
GlobalAvgPool — used before the classification head in modern networks (e.g.: ResNet); produces one vector per channel without

Part 5 — CNN Architecture

Combining convolutions, pooling and dense layers.

Convolutional blocks

A CNN is built from stacked **convolutional blocks** — each block has **greater depth** and **lower resolution** than the previous one — followed by dense layers:



Stacking layers — hierarchy of patterns

Why stack convolutional blocks?

- Each layer sees a **local window** of the previous layer's output
- Deeper layers have a **larger receptive field** — they see more of the image
- The network learns a **hierarchy of representations**, from simple to complex

Empirical visualization (Zeiler & Fergus, 2014): filters of trained networks show a clear progression — early layers detect edges and gradients, middle layers detect textures and parts, deep layers detect complete objects.

Layer 1 → edges and gradients
/ \ - | /
Layer 2 → textures and corners
◻ ▣ ◆ ▨
Layer 3 → object parts
👁 🍷 🦴
Layer 4 → objects / concepts
🐱 🚗 🌸

Depth → increasing abstraction.

Transfer learning works because these representations **generalize** to new domains.

CNN for Fashion-MNIST

Keras

```
inp = keras.Input(shape=(28, 28, 1))
x = keras.layers.Conv2D(
    32, 3, activation='relu', padding='same')(inp)
x = keras.layers.MaxPooling2D()(x)
x = keras.layers.Conv2D(
    64, 3, activation='relu', padding='same')(x)
x = keras.layers.MaxPooling2D()(x)
x = keras.layers.Flatten()(x)
x = keras.layers.Dense(128, activation='relu')(x)
x = keras.layers.Dropout(0.3)(x)
out = keras.layers.Dense(10, activation='softmax')(x)
model = keras.Model(inp, out)
model.compile(
    loss='sparse_categorical_crossentropy',
    optimizer='adam', metrics=['accuracy'])
```

PyTorch

```
model = nn.Sequential(
    nn.Conv2d(1, 32, 3, padding=1), nn.ReLU(),
    nn.MaxPool2d(2),
    nn.Conv2d(32, 64, 3, padding=1), nn.ReLU(),
    nn.MaxPool2d(2),
    nn.Flatten(),
    nn.Linear(64 * 7 * 7, 128), nn.ReLU(),
    nn.Dropout(0.3),
    nn.Linear(128, 10),
)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters())
```

CNN achieves **~92%** on Fashion-MNIST, versus ~88% for the dense network.

Residual Connections (*Skip Connections*)

The problem with very deep networks

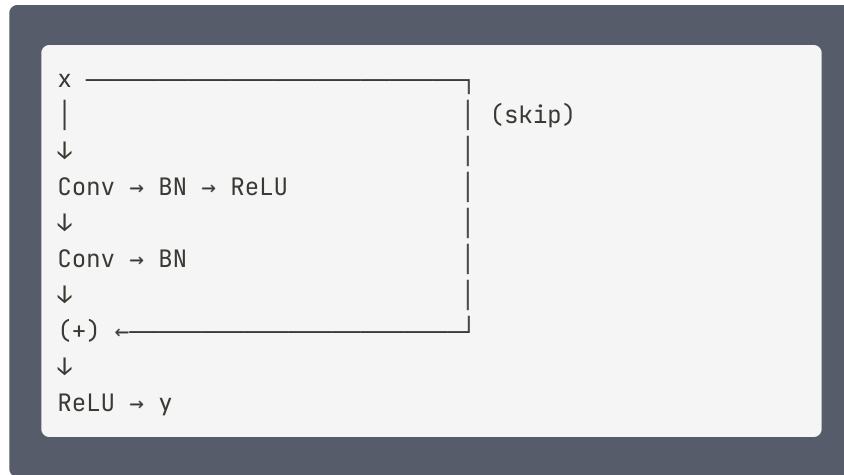
- Gradient dissipates layer by layer (*vanishing gradient*)
- Networks with >20 layers converged worse than shallow ones

Idea (He et al., ResNet 2015): allow the signal to flow directly by skipping layers.

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, W) + \mathbf{x}$$

Instead of learning $\mathbf{y}$, the block learns the **residual** $\mathcal{F} = \mathbf{y} - \mathbf{x}$.

If $\mathcal{F} \approx 0$, the block acts as an **identity** — never makes the previous result worse.



BN = *Batch Normalization*: normalizes the activations of each layer (mean 0, variance 1 per batch), speeding up convergence and stabilizing training.

$$y = \mathcal{F}(\mathbf{x}, W) + \mathbf{x}$$

Gradient flows through the skip without depending on $\mathcal{F} \rightarrow$ ResNet-152 trains successfully.

Part 6 — Transfer Learning

Reusing what was learned from millions of images.

Two trends that enabled transfer learning

Trend 1 — Specialized architectures:

Data type	Architecture
Any	Residual connections (ResNet)
Images	Convolutional layers
Sequences (text, audio, genes)	Transformers

Trend 2 — Publicly available pre-trained models:

Using these architectures, researchers trained high-performance networks on large real-world datasets. Countless models are publicly available.

The core idea of transfer learning

- **ImageNet:** millions of images from 1000 everyday categories
- A network trained on ImageNet develops a **hierarchical** representation of visual objects:
 - Early layers → edges and textures
 - Middle layers → object parts
 - Final layers → complete objects

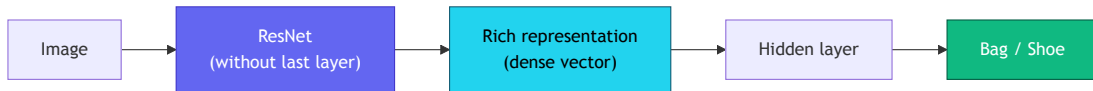
Problem: we only have ~100 images of bags and shoes.
A CNN trained from scratch will not generalize.

Solution: take a ResNet trained on ImageNet and **reuse** what it already learned about everyday images.

Transfer learning = **customizing** a pre-trained network for a new problem.

"Headless" ResNet as a feature extractor

1. The original ResNet classifies 1000 categories — the last layer is not useful to us
2. We remove the last layer ("**headless**" ResNet)
3. We pass our images → the output is a rich, dense representation
4. We train a small network on top of that representation



With only ~100 images, this approach already yields high accuracy.

Feature extraction vs. Fine-tuning

Feature extraction

- Pre-trained base weights remain **frozen**
- Only the new head is trained
- Faster, works with little data
- Recommended when the target dataset is small and similar to ImageNet

Fine-tuning

- Connect base + head and train **everything end-to-end**
- Must start from pre-trained weights (not from scratch)
- Uses a smaller learning rate to avoid "forgetting"
- Better when there is sufficient data

Catastrophic Forgetting

The problem

- During fine-tuning, gradients from the new task **overwrite** weights important for the original task
- The network "forgets" what it learned during pre-training — loses the generic feature representation
- The higher the learning rate and the more layers unfrozen, the more severe the forgetting

Symptom: accuracy on the target dataset rises quickly, but the model loses generalization capacity — especially with limited data.

Mitigation strategies

Low learning rate

Fine-tune with LR 10–100× smaller than original pre-training (e.g. $1e-4$ instead of $1e-2$)

Gradual unfreezing

Unfreeze layers top-to-bottom, one at a time, over the course of training

Discriminative learning rates

Layers near the input get a smaller LR; layers in the head get a larger LR

Feature extraction first

Train only the head until convergence; then unfreeze for fine-tuning

Transfer learning in code

Keras

```
base = keras.applications.ResNet50(
    weights='imagenet',
    include_top=False,
    input_shape=(224, 224, 3),
)
base.trainable = False # freeze

inp = keras.Input(shape=(224, 224, 3))
x = base(inp, training=False)
x = keras.layers.GlobalAveragePooling2D()(x)
x = keras.layers.Dense(64, activation='relu')(x)
out = keras.layers.Dense(2, activation='softmax')(x)

model = keras.Model(inp, out)
model.compile(
    loss='sparse_categorical_crossentropy',
    optimizer='adam', metrics=['accuracy'])
```

PyTorch

```
import torchvision.models as models

base = models.resnet50(weights='IMAGENET1K_V2')

# Freeze all weights
for p in base.parameters():
    p.requires_grad = False

# Replace the classification head
n_feat = base.fc.in_features
base.fc = nn.Sequential(
    nn.Linear(n_feat, 64),
    nn.ReLU(),
    nn.Linear(64, 2),
)

optimizer = optim.Adam(base.fc.parameters())
criterion = nn.CrossEntropyLoss()
```

Where to find pre-trained models



TensorFlow Hub

tensorflow.org/hub

Ready-to-use Keras models



PyTorch Hub

pytorch.org/hub

ResNet, EfficientNet, BERT...



Hugging Face Hub

huggingface.co/models

+500k models: vision, NLP, audio

Part 7 — Self-Supervised Learning

Why pre-train without labels?

The problem

- Labeling data is **expensive and slow**
- Annotated datasets: ImageNet (1.2 M), CIFAR-10 (60 K)
- Unlabeled data: **the entire internet**

The idea

- Train the backbone on an **auxiliary task** created from the data itself
- No human; the label is generated automatically
- The learned backbone is reused (feature extraction / fine-tuning) with few labeled examples

Self-Supervised Learning (SSL) = supervision comes from the data itself, not from human annotators.

The auxiliary task is called a **pretext task**.

Pretext Tasks

A pretext task automatically creates **(input, label)** pairs, forcing the backbone to learn useful representations.

Rotation Prediction

Rotates the image by $0^\circ/90^\circ/180^\circ/270^\circ$.
Predicts which rotation was applied.

Gidaris et al., 2018 — RotNet

Shuffled Patches

Divides into 9 patches and shuffles them.
Predicts the original permutation.

Noroozi & Favaro, 2016 — Jigsaw

Masked Reconstruction

Masks ~75% of patches. Reconstructs the masked pixels.

He et al., 2022 — MAE

Contrastive Learning

Maximizes similarity between views of the same image; minimizes between different ones.

Chen et al., 2020 — SimCLR

Colorization

Receives a grayscale image. Predicts the colors (channels a and b of Lab color space).

Zhang et al., 2016

Context Prediction

Extracts two random patches. Predicts the relative position between them (8 directions).

Doersch et al., 2015

Denoising Autoencoder

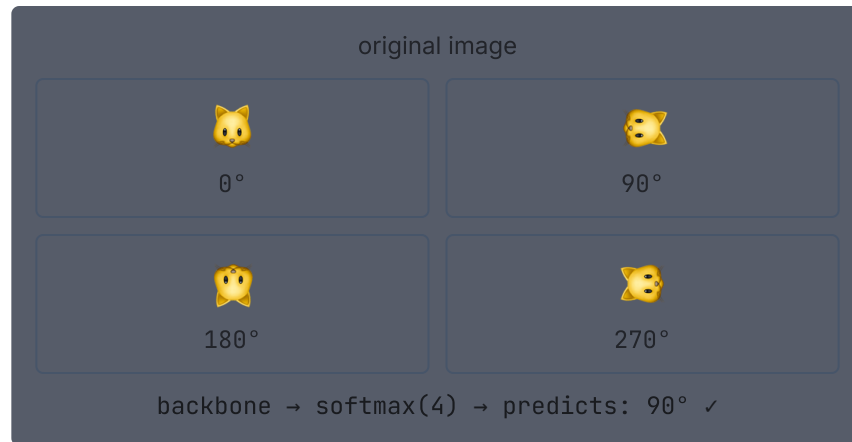
Corrupts the input with noise. Reconstructs the original clean image.

Vincent et al., 2008 — DAE

Rotation Prediction (RotNet)

How it works

1. Takes any image (without label)
2. Applies one of 4 rotations: 0° , 90° , 180° , 270°
3. Backbone extracts features; classification head predicts which rotation (4 classes)
4. Loss: standard Cross-Entropy



Why does it work? To get the rotation right, the model must understand *what is in the image* — the natural orientation of objects. This forces the backbone to learn semantic features.

Shuffled Patches (Jigsaw)

Procedure

1. Divides the image into a 3×3 grid (9 patches)
2. Shuffles the patches with a random permutation π
3. Backbone processes each patch; head predicts π from a pre-defined set
4. Backbone learns spatial relationships and semantic context

Original:

1	2	3
4	5	6
7	8	9

→

Shuffled:

7	4	2
1	9	5
3	6	8

model must predict: $\pi = [7, 4, 2, 1, 9, 5, 3, 6, 8]$

Intuition: To reassemble the image, the model must understand **what belongs together** — a key concept for object recognition.

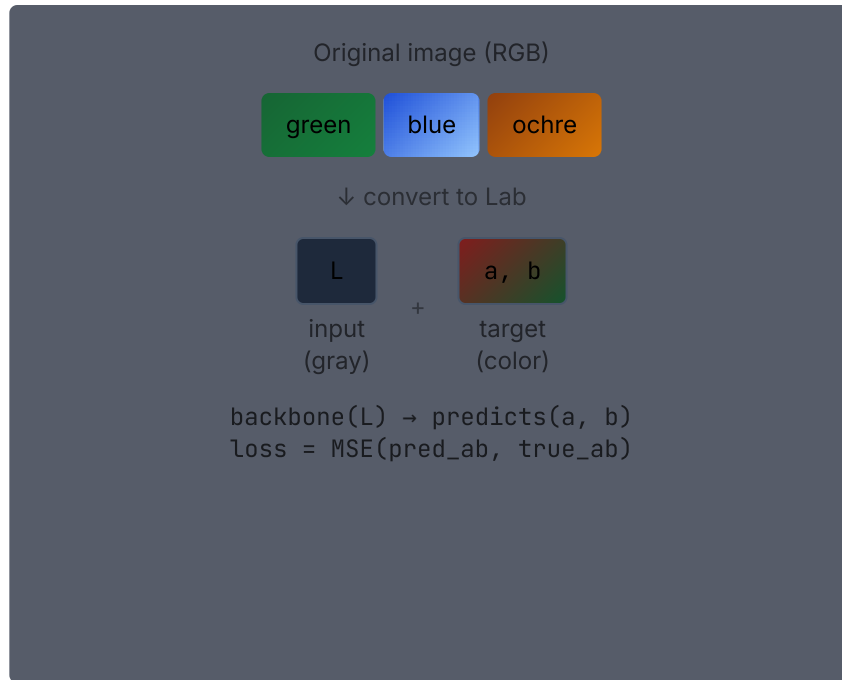
Colorization

Procedure

1. Converts the image to the **Lab** color space
 - Channel **L** = luminosity (grayscale)
 - Channels **a**, **b** = chromaticity (color)
2. Provides only the **L** channel to the backbone
3. Backbone predicts the **a** and **b** channels
4. Loss: mean squared error on the color channels

Why does it work? To colorize correctly, the model must identify *what* each object is — grass is green, sky is blue, skin has specific tones. This forces semantic learning.

Advantage: label comes from the original image — no human annotation needed. Any color photo can serve as training data.



Limitation: color ambiguity (cars can be any color). Models learn to predict "safe" average colors; recent versions use probabilistic distributions.

Context Prediction (*Relative Patch Position*)

Procedure (Doersch et al., 2015)

1. Extracts the **central** patch of the image
2. Extracts a **neighboring** patch from one of the 8 surrounding positions
3. Backbone processes both patches
4. Head predicts which of the **8 directions** the neighbor occupies
5. Loss: Cross-Entropy (8 classes)

Why does it work? To get the relative position right, the model must understand object parts and their spatial relations — e.g., eyes are above the nose, wheels are below the car.

Possible positions (8 directions)



```
backbone(patch_ref, patch_neighbor) → softmax(8)  
predicts: position 2 (above) ✓
```

Masked Autoencoders (MAE)

Idea (He et al., 2022)

- Divides the image into 16×16 patches
- Masks ~75% of patches randomly
- **Encoder** (ViT) processes only visible patches
- Lightweight **decoder** reconstructs the masked pixels
- Loss: MSE on masked pixels

input image (patches):

■ ■ ■ ■ ■ ■ ■ ■

■ ■ ■ ■ ■ ■ ■ ■ (64 patches)

masking (75%):

⊞ ■ ⊞ ⊞ ■ ⊞ ■ ⊞

⊞ ⊞ ■ ⊞ ⊞ ■ ⊞ ⊞ ⊞ = masked

Encoder → processes only ■

Decoder → reconstructs ⊞

→ backbone learns dense representation
of the visible parts

Why 75%? Small masks allow trivial interpolation. 75% forces global scene understanding.

Result: MAE pre-trains ViT-L on ImageNet in ~31 h (8 A100 GPUs) and achieves SOTA in classification with fine-tuning.

Contrastive Learning (SimCLR)

Principle

- Creates 2 **augmented views** of the same image (crop, flip, color jitter...)
- Maximizes similarity between views of the **same** image
- Minimizes similarity with **other images** in the batch

NT-Xent Loss (InfoNCE):

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

τ = temperature; z = projected embedding

Large batch $\rightarrow$ more negatives $\rightarrow$ better signal. SimCLR uses batch size 4096.

```
image x
├─ aug1 → encoder → g(·) → z1
└─ aug2 → encoder → g(·) → z2 ┘ → max sim(z1, z2)
```

other images in batch:

```
x' → encoder → g(·) → z'
                                → min sim(z1, z')
```

encoder: shared weights
g(·): projector, discarded
after pre-training

Denoising Autoencoder (DAE)

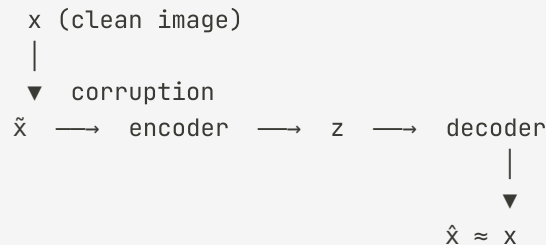
Core idea (Vincent et al., 2008)

- Corrupts the input: Gaussian noise, *salt-and-pepper*, region masking
- The model learns to map the corrupted version $\tilde{x}$ back to the original x
- The encoder is forced to capture semantic structure — it cannot memorize noise

Reconstruction loss:

$$\mathcal{L} = \|x - \text{dec}(\text{enc}(\tilde{x}))\|^2$$

Difference from MAE: DAE uses continuous noise on pixels; MAE masks entire patches. MAE scales better for ViTs.



noise types:

- gaussian: $\tilde{x} = x + \varepsilon, \varepsilon \sim N(0, \sigma^2)$
- masking: pixels $\rightarrow 0$ with prob. p
- salt-pepper: pixels $\rightarrow 0$ or 255

SSL Methods Comparison

Method	Signal type	Architecture	Strengths
RotNet	Classification (4 classes)	CNN	Simple; good for quick pretraining
Jigsaw	Permutation classification	CNN	Learns spatial relationships
Colorization	Regression (Lab colors)	CNN encoder-decoder	Learns semantics of natural objects
Context Pred.	Classification (8 directions)	CNN	Learns spatial scene layout
DAE	Reconstruction (noise → clean)	CNN encoder-decoder	Robust to corruptions; historical pioneer
MAE	Pixel reconstruction	ViT + decoder	SOTA on ViTs; efficient
SimCLR	Contrastive (similarity)	CNN/ViT	Label-free; very flexible
DINO	Self-distillation	ViT	Emergent segmentation features

Generative methods (MAE, Jigsaw): the model reconstructs/reorders data; learns detailed geometry and texture.

Contrastive methods (SimCLR, DINO): the model learns invariances; learns high-level semantics.

Summary

Tensor $H \times W \times 3$ — representation of color images; batch: $N \times H \times W \times 3$

Softmax + Cross-Entropy — multiclass output; correct loss for N categories

Flatten + Dense — simple baseline, but explodes in parameters and loses spatial structure

Convolutional filter — local window with shared weights; learns once, applies across the entire image

Max Pooling — downsampling that preserves feature presence; robustness to shifts

CNN — Conv+ReLU+Pool blocks; depth grows, resolution shrinks

Feature extraction — frozen base + new head; works with little data

Fine-tuning — trains everything end-to-end starting from pre-trained weights

Pretext task — automatically generated label from the data; forces the backbone to learn useful representations

SSL — RotNet, Jigsaw, MAE, SimCLR; pre-training without human annotations

Next lecture

- **Sequences and language** — how to represent text for neural networks
- **Embeddings** — dense vectors for words and tokens
- **RNNs and LSTMs** — networks that process sequences step by step
- **Attention and Transformers** — the mechanism that dominates modern NLP
- **BERT and GPT** — pre-training and fine-tuning for language tasks

Thank you! Questions?

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